

# Principles of Electronic Communication Lab

Hands-on laboratory course for communication components, AM/FM generator and demodulator circuits, pulse modulation circuits and FM receiver-related stages.

Course Code: 3046

Category: Program Core

Credits: 1.5

L:T:P = 0:0:3

Lab / Practical

## COURSE OVERVIEW

### Principles of Electronic Communication Lab (3046)

This is a standalone digital textbook, revision guide, workbook and answer guide prepared from the uploaded official syllabus PDF.

### Student-friendly summary

Hands-on laboratory course for communication components, AM/FM generator and demodulator circuits, pulse modulation circuits and FM receiver-related stages.

**Study tip:** ഓരോ module-ഉം separate ആയി പഠിക്കുക. Definitions, diagrams, formulas/procedures, and expected answers English technical terms ഉപയോഗിച്ച് practise ചെയ്യുക.

## Course Metadata

|                               |                                                                  |
|-------------------------------|------------------------------------------------------------------|
| <b>Course Code</b>            | 3046                                                             |
| <b>Base Course Code</b>       | 3046                                                             |
| <b>Course Title</b>           | Principles of Electronic Communication Lab                       |
| <b>Programme / Department</b> | Diploma in Electronics / Electronics & Communication Engineering |
| <b>Semester</b>               | 3                                                                |
| <b>Course Category</b>        | Program Core                                                     |
| <b>Credits</b>                | 1.5                                                              |
| <b>L:T:P</b>                  | 0:0:3                                                            |
| <b>Periods / semester</b>     | 45                                                               |
| <b>Subject Type</b>           | Lab / Practical                                                  |

## Course Objectives

- Give hands-on experience with electronic components, analogue modulation circuits and pulse modulation circuits.
- Demonstrate the working of various stages of FM and solve associated problems.
- Guide students in developing an FM receiver of high fidelity.

## Prerequisite Knowledge

- Testing active/passive components and equipment familiarization - Fundamentals of Electrical and Electronics Engineering Lab
- Operation of transistor and diode - Basic Electronics

## Course Outcomes

|      |                                                                      |    |               |
|------|----------------------------------------------------------------------|----|---------------|
| CO 1 | Illustrate working of electronic components in communication system. | 3  | Understanding |
| CO 2 | Experiment with analog modulation schemes.                           | 12 | Applying      |
| CO 3 | Develop pulse modulation circuits.                                   | 9  | Applying      |
| CO 4 | Build different stages in AM and FM.                                 | 15 | Applying      |

## Module Map

1. **Module 1:** Communication components familiarization (3 h)
2. **Module 2:** Analog modulation experiments (12 h)
3. **Module 3:** Pulse modulation circuits (9 h)
4. **Module 4:** AM/FM stages and open-ended FM receiver (15 h)

## How to study this subject

- Read the aim and apparatus before lab.
- Draw the circuit/setup diagram before wiring.
- Write procedure and observation table in advance.
- After experiment, write result and precautions immediately.
- Practise viva questions module-wise.

## Exam preparation method

- Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.
- Practise neat wiring/setup explanation.
- Prepare safety precautions and viva answers.
- Know how to calculate/read final result from observations.

## Quick Navigation Cards

# Experiment Bank - Principles of Electronic Communication Lab

Quick reference cards for revision. Use this only after reading the module explanation.

## Record format

**Aim -> Apparatus -> Circuit -> Procedure -> Observation -> Result**

**Meaning:** Standard practical record structure.

**Where used:** All lab experiments.

**Common mistake:** Leaving observation table incomplete.

## AM index from envelope

**$m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$**

**Meaning:** Find modulation index from CRO waveform.

**Where used:** AM generator experiment.

**Common mistake:** Measuring  $V_{\max}/V_{\min}$  from wrong points.

## 555 IC role

**Timer / pulse generator**

**Meaning:** Used for FM/PWM/PPM circuits.

**Where used:** Pulse modulation experiments.

**Common mistake:** Wrong power pins or reset pin.

## 565 IC role

**Meaning:** Used for FM demodulation/frequency multiplier.

**Where used:** FM generator/demodulator and multiplier.

**Common mistake:** Not setting lock range correctly.

## FM receiver band

**80 MHz - 108 MHz**

**Meaning:** Open-ended FM receiver setup range.

**Where used:** Open-ended experiment.

**Common mistake:** Wrong antenna/tuning range.

MODULE 1 • 3 HOURS

## Communication components familiarization

Familiarize 555, 565, LM380 and counter IC 7490 used in communication circuits.

## Learning outcome

Familiarize 555, 565, LM380 and counter IC 7490 used in communication circuits.

**Malayalam support:** ആദ്യം meaning മനസ്സിലാക്കുക, പിന്നെ English technical terms ഉപയോഗിച്ച് answer എഴുതുക.

## Important terms

- 555 timer IC
- 565 PLL IC
- LM380 audio amplifier
- Counter IC 7490
- Pin identification and datasheet reading

M1.01

Familiarize 555, 565, LM380 and Counter IC 7490

3

Understanding

## Topic-wise explanation

**555 timer IC:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**565 PLL IC:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**LM380 audio amplifier:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Counter IC 7490:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Pin identification and datasheet reading:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

## Visual block / diagram idea



Communication components familiarization: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

## Worked / practice example

**Practical example:** For AM generation, set carrier and modulating signals, observe output envelope, measure  $V_{\max}$  and  $V_{\min}$ , and calculate modulation index  $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$ .

## Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

## Expected Questions

**M1-Q1** Write the aim and apparatus required for an experiment based on Communication components familiarization.

**M1-Q2** Explain the step-by-step procedure for Communication components familiarization.

**M1-Q3** Draw or describe the required circuit/setup and observation table for Communication components familiarization.

**M1-Q4** List important precautions and safety points while doing Communication components familiarization.

**M1-Q5** Write viva questions with answers for Communication components familiarization.

**M1-Q6** How will you write the final result and verify correctness in Communication components familiarization?

## Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

### MODULE 2 • 12 HOURS

## Analog modulation experiments

Construct and observe AM/FM generator and demodulator circuits.

## Learning outcome

Construct and observe AM/FM generator and demodulator circuits.

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## Important terms

AM generator using transistor

Modulation index measurement

AM demodulator using diode

Input-output waveform observation



## Official module outcome table

| SYLLABUS CODE | MODULE OUTCOME                                                       | HOURS | LEVEL    |
|---------------|----------------------------------------------------------------------|-------|----------|
| M2.01         | Construct AM generator using transistor and measure modulation index | 3     | Applying |
| M2.02         | Construct AM demodulator using diode and observe waveforms           | 3     | Applying |
| M2.03         | Build FM generator using 555 IC and observe waveforms                | 3     | Applying |
| M2.04         | Build FM generator and demodulator using 565 IC                      | 3     | Applying |

## Topic-wise explanation

**AM generator using transistor:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Modulation index measurement:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**AM demodulator using diode:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Input-output waveform observation:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**FM generator using 555 IC:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**FM generator and demodulator using 565 IC:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

## Visual block / diagram idea



Analog modulation experiments: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

## Worked / practice example

**Practical example:** For AM generation, set carrier and modulating signals, observe output envelope, measure  $V_{\max}$  and  $V_{\min}$ , and calculate modulation index  $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$ .

## Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

## Expected Questions

**M2-Q1** Write the aim and apparatus required for an experiment based on Analog modulation experiments.

**M2-Q2** Explain the step-by-step procedure for Analog modulation experiments.

**M2-Q3** Draw or describe the required circuit/setup and observation table for Analog modulation experiments.

**M2-Q4** List important precautions and safety points while doing Analog modulation experiments.

**M2-Q5** Write viva questions with answers for Analog modulation experiments.

**M2-Q6** How will you write the final result and verify correctness in Analog modulation experiments?

## Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

### MODULE 3 • 9 HOURS

## Pulse modulation circuits

Develop PAM, PWM and PPM modulator/demodulator circuits and observe waveforms.

## Learning outcome

Develop PAM, PWM and PPM modulator/demodulator circuits and observe waveforms.

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# Important terms

PAM modulator and demodulator

PWM modulator using 555 IC

PPM modulator using 555

Input-output waveform comparison

Duty cycle and pulse position observation

## Official module outcome table

| SYLLABUS CODE | MODULE OUTCOME                                | HOURS | LEVEL    |
|---------------|-----------------------------------------------|-------|----------|
| M3.01         | Develop PAM modulator and demodulator circuit | 3     | Applying |
| M3.02         | Develop PWM modulator using 555 IC            | 3     | Applying |
| M3.03         | Develop PPM modulator using 555               | 3     | Applying |

## Topic-wise explanation

**PAM modulator and demodulator:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**PWM modulator using 555 IC:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**PPM modulator using 555:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Input-output waveform comparison:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Duty cycle and pulse position observation:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

## Visual block / diagram idea



Pulse modulation circuits: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

## Worked / practice example

**Practical example:** For AM generation, set carrier and modulating signals, observe output envelope, measure  $V_{\max}$  and  $V_{\min}$ , and calculate modulation index  $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$ .

## Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

## Expected Questions

**M3-Q1** Write the aim and apparatus required for an experiment based on Pulse modulation circuits.

**M3-Q2** Explain the step-by-step procedure for Pulse modulation circuits.

**M3-Q3** Draw or describe the required circuit/setup and observation table for Pulse modulation circuits.

**M3-Q4** List important precautions and safety points while doing Pulse modulation circuits.

**M3-Q5** Write viva questions with answers for Pulse modulation circuits.

**M3-Q6** How will you write the final result and verify correctness in Pulse modulation circuits?

## Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

### MODULE 4 • 15 HOURS

## AM/FM stages and open-ended FM receiver

Build pre-emphasis, de-emphasis, frequency multiplier, mixer, audio amplifier and FM receiver setup.

## Learning outcome

Build pre-emphasis, de-emphasis, frequency multiplier, mixer, audio amplifier and FM receiver setup.

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## Important terms

Pre-emphasis circuit

De-emphasis circuit

Frequency multiplier using IC 565

## Official module outcome table

| SYLLABUS CODE | MODULE OUTCOME                                                       | HOURS | LEVEL    |
|---------------|----------------------------------------------------------------------|-------|----------|
| M4.01         | Build pre-emphasis and de-emphasis circuits and plot characteristics | 3     | Applying |
| M4.02         | Construct frequency multiplier using IC 565                          | 3     | Applying |
| M4.03         | Build mixer using discrete components                                | 3     | Applying |
| M4.04         | Build audio power amplifier circuit                                  | 3     | Applying |
| Open-ended    | Setup FM receiver in range 80 MHz - 108 MHz                          | 3     | Applying |

## Topic-wise explanation

**Pre-emphasis circuit:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**De-emphasis circuit:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Frequency multiplier using IC 565:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Mixer circuit using discrete components:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**Audio power amplifier circuit:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

**FM receiver setup in 80 MHz-108 MHz range:** Understand the aim, identify the required IC/components/instruments, wire the circuit or setup carefully, observe the waveform/output, record readings, and write the result with precautions.

## Visual block / diagram idea



AM/FM stages and open-ended FM receiver: lab record and practical workflow

Use this type of labelled diagram/block in answers when applicable.

## Worked / practice example

**Practical example:** For AM generation, set carrier and modulating signals, observe output envelope, measure  $V_{\max}$  and  $V_{\min}$ , and calculate modulation index  $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$ .

## Common mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.



## Expected Questions

**M4-Q1** Write the aim and apparatus required for an experiment based on AM/FM stages and open-ended FM receiver.

**M4-Q2** Explain the step-by-step procedure for AM/FM stages and open-ended FM receiver.

**M4-Q3** Draw or describe the required circuit/setup and observation table for AM/FM stages and open-ended FM receiver.

**M4-Q4** List important precautions and safety points while doing AM/FM stages and open-ended FM receiver.

**M4-Q5** Write viva questions with answers for AM/FM stages and open-ended FM receiver.

**M4-Q6** How will you write the final result and verify correctness in AM/FM stages and open-ended FM receiver?

## Module checklist

- ✓ Aim and apparatus are clear.
- ✓ Circuit/setup is neat and labelled.
- ✓ Procedure is in correct order.
- ✓ Observation table and result are complete.
- ✓ Precautions and viva points are prepared.

### QUICK REVISION

## One-page Revision Summary

Use this page in the last 24 hours before exam or lab evaluation.

## Module-wise must-know points

1. **Module 1 - Communication components familiarization:** Know IC purpose, pins, ratings and typical communication applications.
2. **Module 2 - Analog modulation experiments:** Prepare AM/FM experiment aims, circuits, waveforms, modulation index and precautions.
3. **Module 3 - Pulse modulation circuits:** Know difference between PAM, PWM and PPM with waveform observations.
4. **Module 4 - AM/FM stages and open-ended FM receiver:** Revise circuit purpose, procedure, observations and open-ended FM receiver report.

## Important definitions / keywords

AM generator

AM demodulator

FM generator

PLL

PAM

PWM

PPM

pre-emphasis

de-emphasis

mixer

audio power amplifier

FM receiver

## Important formulas / key points

- Aim -> Apparatus -> Circuit -> Procedure -> Observation -> Result
- $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$
- Timer / pulse generator
- PLL IC
- 80 MHz - 108 MHz

## Common exam mistakes

- Wiring the circuit without switching off the supply.
- Using wrong IC pin number or wrong instrument range.
- Not drawing waveform axes and scale clearly.
- Writing result without observation or calculation.
- Ignoring component polarity and maximum ratings.

## Last-minute checklist

- ✓ Memorise lab record format: aim, apparatus, theory, diagram, procedure, observation, result.

- ✓ Practise neat wiring/setup explanation.  
Course 3046
- ✓ Prepare safety precautions and viva answers.
- ✓ Know how to calculate/read final result from observations.

## Diagram / table list

- AM generator circuit
- AM diode demodulator circuit
- FM generator using 555
- FM generator/demodulator using 565
- PAM/PWM/PPM circuits
- Pre-emphasis/de-emphasis circuits
- Frequency multiplier block
- Mixer circuit
- Audio power amplifier circuit
- FM receiver setup block

### MASTER ANSWER KEY

## Master Answer Key - matched with Expected Questions

Every answer below matches the module expected questions exactly. Attempt first, then verify here.

### M1-Q1 - Write the aim and apparatus required for an experiment based on Communication components familiarization.

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M1-Q2 - Explain the step-by-step procedure for Communication components familiarization.

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M1-Q3 - Draw or describe the required circuit/setup and observation table for Communication components familiarization.

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M1-Q4 - List important precautions and safety points while doing Communication components familiarization.

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M1-Q5 - Write viva questions with answers for Communication components familiarization.

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M1-Q6 - How will you write the final result and verify correctness in Communication components familiarization?

**Aim:** Perform the experiment related to Communication components familiarization and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M2-Q1 - Write the aim and apparatus required for an experiment based on Analog modulation experiments.

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M2-Q2 - Explain the step-by-step procedure for Analog modulation experiments.

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply,

function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable.

Course 3046

**Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M2-Q3 - Draw or describe the required circuit/setup and observation table for Analog modulation experiments.

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M2-Q4 - List important precautions and safety points while doing Analog modulation experiments.

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M2-Q5 - Write viva questions with answers for Analog modulation experiments.

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply

input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M2-Q6 - How will you write the final result and verify correctness in Analog modulation experiments?

**Aim:** Perform the experiment related to Analog modulation experiments and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M3-Q1 - Write the aim and apparatus required for an experiment based on Pulse modulation circuits.

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

## M3-Q2 - Explain the step-by-step procedure for Pulse modulation circuits.

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct



polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M3-Q3 - Draw or describe the required circuit/setup and observation table for Pulse modulation circuits.

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M3-Q4 - List important precautions and safety points while doing Pulse modulation circuits.

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M3-Q5 - Write viva questions with answers for Pulse modulation circuits.

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.



### M3-Q6 - How will you write the final result and verify correctness in Pulse modulation circuits?

**Aim:** Perform the experiment related to Pulse modulation circuits and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M4-Q1 - Write the aim and apparatus required for an experiment based on AM/FM stages and open-ended FM receiver.

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M4-Q2 - Explain the step-by-step procedure for AM/FM stages and open-ended FM receiver.

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

### M4-Q3 - Draw or describe the required circuit/setup and observation table for AM/FM stages and open-ended FM receiver.

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

#### **M4-Q4 - List important precautions and safety points while doing AM/FM stages and open-ended FM receiver.**

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

#### **M4-Q5 - Write viva questions with answers for AM/FM stages and open-ended FM receiver.**

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

#### **M4-Q6 - How will you write the final result and verify correctness in AM/FM stages and open-ended FM receiver?**

**Aim:** Perform the experiment related to AM/FM stages and open-ended FM receiver and verify the expected behaviour. **Apparatus:** Use the required trainer kit/components, regulated

supply, function generator, CRO/DSO, multimeter, connecting wires and datasheets as applicable. **Procedure:** Check components, wire the circuit neatly, set correct instrument ranges, apply input gradually, take readings, plot waveform/characteristics, and compare with expected result. **Precautions:** Switch off supply before wiring, avoid loose connections, use correct polarity and never exceed ratings. **Result:** State the observed result clearly with waveform, table, calculation or program output as required.

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